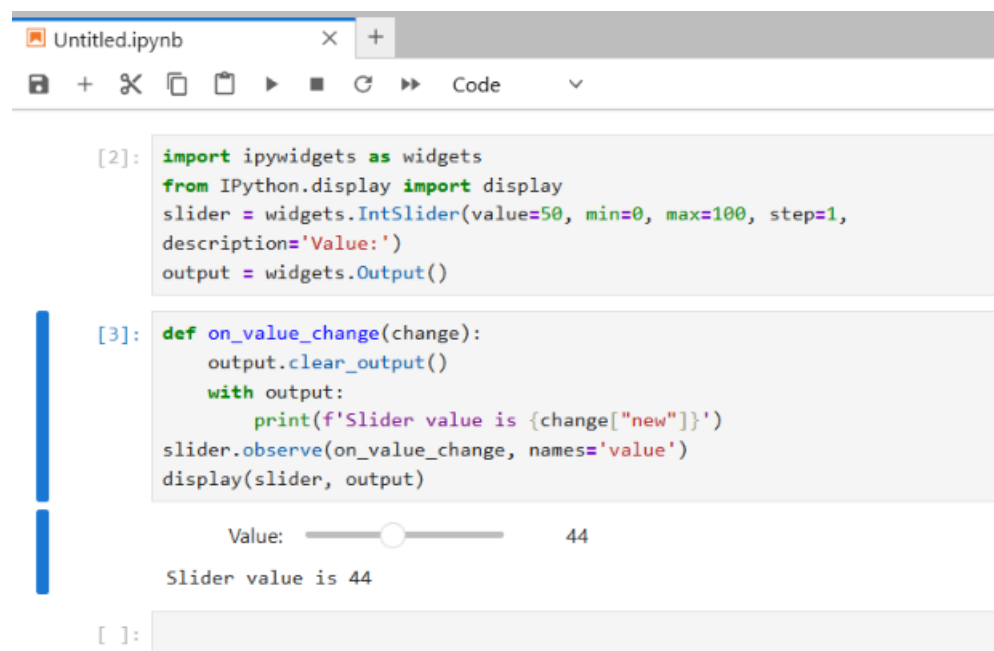


Expt 1

Setting up the Python environment and libraries

Aim:

- Create a new notebook for Python
- Write and execute Python code
- Create new cells for code and Markdown
- Demonstrate the application of Jupyter Widgets



The screenshot shows a Jupyter Notebook window titled 'Untitled.ipynb'. The interface includes a toolbar with icons for saving, adding, deleting, and running code. The first code cell (index [2]) contains the following Python code:

```
[2]: import ipywidgets as widgets
      from IPython.display import display
      slider = widgets.IntSlider(value=50, min=0, max=100, step=1,
                                description='Value:')
      output = widgets.Output()
```

The second code cell (index [3]) contains the following Python code:

```
[3]: def on_value_change(change):
      output.clear_output()
      with output:
          print(f'Slider value is {change["new"]}')
      slider.observe(on_value_change, names='value')
      display(slider, output)
```

The output of the second cell shows a slider widget with the label 'Value:' and a value of 44. Below the slider, the text 'Slider value is 44' is displayed.

Result:

Value: 44
Slider value is 44